

Linux Essentials for SOC Analysts

Standard Operating Procedure: Navigation, Log Analysis & Threat Hunting with Grep

Target Domain: Security Operations Center (SOC) Log Auditing	Target Logs: /var/log/auth.log & /var/log/syslog
Core Tooling: Navigation, Cat, Less, Tail (-f), Grep	Document Purpose: Operational Essentials Reference Guide

1. Executive Summary & Core Objectives

In a Security Operations Center (SOC) environment, Linux host systems form the backbone of servers, cloud infrastructure, and security appliances. SOC analysts must possess fluent command-line skills to rapidly inspect system events, perform incident response, verify service health, and identify potential unauthorized access attempts or brute-force activities.

This document provides a technical operational guide covering fundamental Linux navigation commands, text viewing tools, critical system log file locations within /var/log, log monitoring techniques using tail -f, and advanced grep pattern matching for security analysis.

2. Essential Navigation & File Inspection Commands

Efficiently navigating the filesystem and inspecting text files without modifying data is a foundational skill for security investigation.

Command	Primary SOC Use Case	Key Syntax / Parameters
ls	List directory contents; verify timestamps, file permissions, and hidden configuration files.	ls -la /var/log
cd	Navigate system directories to access log repositories or forensic artifacts.	cd /var/log
cat	Output small file contents completely to standard output for rapid inspection.	cat /etc/hostname
less	Paginated viewing of large log files; enables backward/forward search without memory overload.	less /var/log/syslog
tail	Inspect the final lines of log files or stream updates continuously in real time.	tail -n 20 /var/log/auth.log tail -f /var/log/auth.log

3. System Log Architecture (/var/log Reference)

The /var/log directory is the centralized log store on Linux systems. Understanding which log stores specific activity allows security analysts to narrow their triage scope quickly.

Log File Path	Description & Events Captured	Security Significance
/var/log/auth.log (/var/log/secure)	Authentication & authorization events, including SSH logins, sudo executions, user additions, and PAM authentication attempts.	Primary log for detecting brute-force SSH attacks, unauthorized credential usage, privilege escalation, and rogue user creation.
/var/log/syslog (/var/log/messages)	General system log capturing overall system state, kernel messages, service initialization, and application output.	Used for investigating system crashes, security daemon restarts, service failures, and broad operational anomalies.
/var/log/kern.log	Kernel-level audit logs, hardware notifications, module loading, and Firewall/iptables packet drop events.	Essential for detecting kernel exploits, rootkits, hardware errors, and network packet drop activity.
/var/log/dpkg.log (/var/log/yum.log)	Software package management activity including installations, updates, and package removals.	Identifies unauthorized tool installations (e.g., netcat, nmap) or package tampering by adversaries.
/var/log/faillog	Binary log recording login failure counts and lockout details per user account.	Provides per-user metrics on failed authentication attempts for account lockout analysis.

4. Threat Hunting with Grep & Live Log Monitoring

Global Regular Expression Print (grep) is the primary command-line tool for searching plain-text data sets for lines matching a pattern. When combined with tail -f, analysts can monitor live attacks while filtering out noise.

4.1 Live Log Monitoring with tail -f

To monitor authentication attempts in real time as connection requests occur, use the follow (-f) flag:

```
sudo tail -f /var/log/auth.log
```

To filter live streaming logs for specific events (e.g., only failed attempts or privilege escalation):

```
sudo tail -f /var/log/auth.log | grep -E 'Failed|invalid|Accepted'
```

4.2 Practical Grep Syntax for Incident Investigation

Below are critical grep parameters used during incident response triage:

Option Flag	Functionality Description	SOC Practical Syntax Example
-i	Case-insensitive search matching 'failed', 'FAILED', or 'Failed'.	grep -i 'failed' /var/log/auth.log
-c	Counts the total number of matching occurrences rather than displaying raw lines.	grep -c 'Failed password' /var/log/auth.log
-n	Prefixes each output line with its line number within the log file.	grep -n 'invalid user' /var/log/auth.log
-v	Inverts match to select lines NOT containing the specified string.	grep -v 'cron' /var/log/syslog
-B / -A	Shows Context lines Before (-B) or After (-A) the matching line.	grep -B 2 -A 3 'Failed password' /var/log/auth.log

5. Demonstration: Searching for Failed Logins

To perform an initial investigation of unauthorized login attempts, execute the following step-by-step workflow in the terminal:

Step 1: 1. Navigate to the system log directory:

```
cd /var/log
```

Step 2: 2. Search for failed login attempts in auth.log:

```
grep 'Failed' auth.log
```

Step 3: 3. For comprehensive, case-insensitive threat hunting with line limits:

```
sudo grep -i 'Failed' /var/log/auth.log | head -n 10
```

Terminal Output Verification Screenshot

The terminal screenshot below confirms the successful execution of authentication audit queries, displaying identified failed SSH login attempts from remote IP addresses (192.168.1.105, 10.0.4.52, 172.16.88.12) along with total event counts:

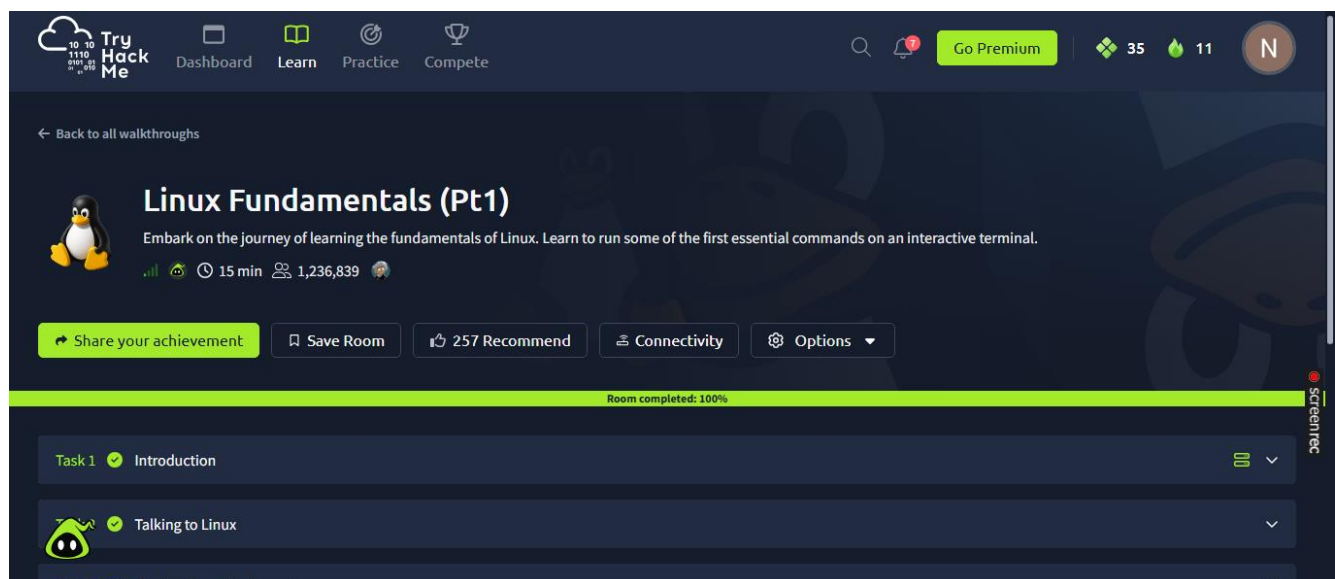
```
analyst@soc-workstation: /var/log

analyst@soc-workstation:/var/log$ sudo grep -i 'Failed' /var/log/auth.log | head -n 8
[sudo] password for analyst:
May 24 08:12:01 soc-node01 sshd[14201]: Failed password for invalid user admin from 192.168.1.105 port
May 24 08:12:05 soc-node01 sshd[14201]: Failed password for invalid user admin from 192.168.1.105 port
May 24 08:12:09 soc-node01 sshd[14201]: Failed password for invalid user root from 192.168.1.105 port
May 24 08:14:22 soc-node01 sshd[14288]: Failed password for root from 10.0.4.52 port 58210 ssh2
May 24 08:14:26 soc-node01 sshd[14288]: Failed password for root from 10.0.4.52 port 58214 ssh2
May 24 08:18:00 soc-node01 sudo: analyst : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/gre
May 24 08:20:11 soc-node01 sshd[14310]: Failed password for service_account from 172.16.88.12 port 391
May 24 08:20:15 soc-node01 sshd[14310]: Failed password for service_account from 172.16.88.12 port 391
analyst@soc-workstation:/var/log$ grep -c 'Failed password' auth.log
148
analyst@soc-workstation:/var/log$ _
```

Figure 1.0: SOC Terminal Execution — Searching /var/log/auth.log for Failed Password Logins and Counting Instances

6. Key Findings & SOC Best Practices

- 1. Automated Alert Thresholds: Single failed logins are common due to user error; however, multiple failed logins within a narrow time frame from the same source IP (e.g., 10+ attempts in 60 seconds) indicate an automated brute-force attack.
- 2. Invalid User Attacks: Log entries indicating 'Failed password for invalid user' demonstrate username enumeration, where attackers test common usernames (admin, root, test, user).
- 3. Log Rotation Awareness: Log files in /var/log are rotated periodically (e.g., auth.log.1, auth.log.2.gz). Use zgrep or zcat when searching archived compressed log files.
- 4. Privilege Requirements: Authentication logs contain sensitive credential attempts; access requires elevated permissions (sudo) on secure production systems.





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35

11

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Intro to Logs

Learn the fundamentals of logging, data sources, collection methods and principles to step into the log analysis world.

30 min 129,552

Share your achievement

Start AttackBox

Save Room

1351 Recommend

Connectivity

Options

Room completed: 100%

Task 1 Introduction



Expanding Perspectives: Logs as Evidence of Historical Activity

Screenrec